Universe creation by tunnelling from laboratory data: no manifold instanton, a unique canonical exponent, and a Smarr identity

Project Baby Universe — Genesis Branch

21 October 2026 (Stage One manuscript, draft)

Abstract

We revisit the thin-wall model of a false-vacuum bubble (de Sitter interior, Schwarzschild exterior) and the proposal of Farhi, Guth and Guven that such a bubble, too small to inflate classically, may tunnel into an inflating, causally disconnected daughter universe. (i) We prove that, for every transition starting from regular, time-symmetric, asymptotically flat ("laboratory") data, the Euclidean exterior interpolating region cannot be embedded in Euclidean Schwarzschild. The pseudomanifold encountered by Farhi–Guth–Guven is therefore forced. The obstruction is tied to a sign change of the exterior extrinsic curvature while the wall recedes from the Euclidean horizon; an interior sign change does not obstruct embedding. (ii) In the reduced canonical theory with the asymptotic clock, the branch of the Euclidean momentum is fixed by analyticity. The competing branch is a horizon monodromy, which in Euclidean language is the half-turn slicing jump. (iii) At $M = 0$ the exponent is $B = S_E[\text{instanton}] + \frac{1}{4G}(\sum A_{\max} - \sum A_{\min})$ in closed form; it equals the Coleman–De Luccia exponent of the false vacuum when the false vacuum is the larger cap. (iv) For $M > 0$ the exponent obeys a first law $d\tilde{B} = (\mathrm{Vol}_E^{\mathrm{in}}/8\pi G)\, d\Lambda_{\mathrm{in}} + A_E^W\, d\sigma - \Delta t_E^W\, dM$ and a Smarr identity $B = -\frac{\Lambda_{\mathrm{in}}}{8\pi G}\mathrm{Vol}_E^{\mathrm{in}} - \frac{\sigma}{2}A_E^W - \frac{M}{2}\Delta t_E^W + \frac{\mathcal{A}}{4G}$. This is a covariant expression in which the pseudomanifold enters only through one boundary term. Numerically, over laboratory transitions $B \geq 0.56\, B(M \to 0)$ for all parameters and $B \geq B_{\mathrm{CdL}}$ when $\chi \gtrsim 1.2\kappa$; for GUT-scale parameters and gram-scale bubbles $B \sim 10^6$. All statements are verified symbolically and numerically with an open-source code.

1 Introduction

A spherical region of false vacuum separated from true vacuum by a domain wall can have an inflating interior even though, seen from outside, it is a black hole [1]. Farhi and Guth showed that such a configuration cannot arise classically from regular initial data when the null energy condition holds [2]. Farhi, Guth and Guven (FGG) asked whether quantum tunnelling could do it [3]. Their canonical calculation was hampered by the choice of slicing, and their Euclidean interpolation was not a manifold but a "pseudomanifold" with a conjectured action. Fischler, Morgan and Polchinski found agreement within a Hamiltonian treatment [4]. Later Hamiltonian work extended these transitions to all signs of the cosmological constant [7, 8]. Holographic arguments suggest that smooth saddles do not produce inflating bubbles [9, 10].

This paper makes the FGG difficulties precise and removes the ambiguity in the exponent within the reduced canonical theory. Section 2 fixes the model and conventions. Section 3 summarises the global structure. Section 4 proves the embedding obstruction. Section 5 fixes the branch of the canonical exponent. Sections 6–7 give the closed form at $M = 0$, the first law and the Smarr identity. Section 8 collects numerical results and Section 9 the limitations.

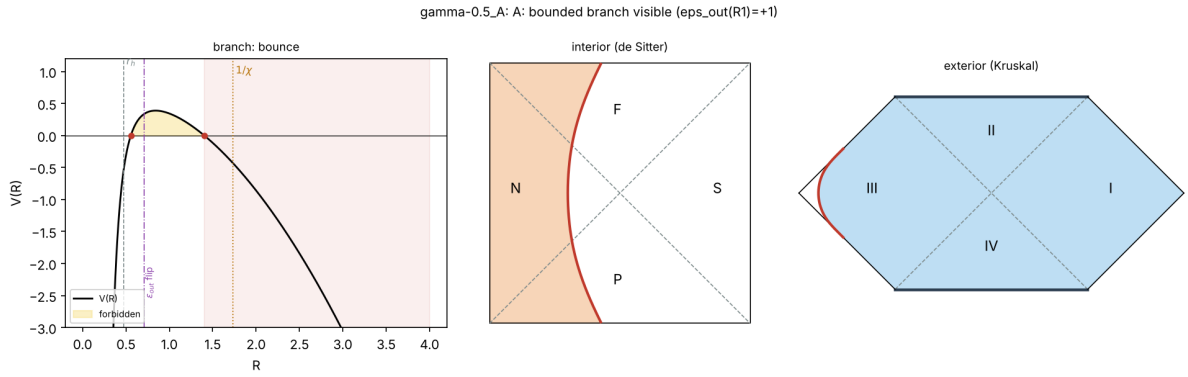


Figure 1: Inflating bounce branch ($\gamma = -0.5$, regime A): effective potential, interior de Sitter diagram, exterior Kruskal diagram, all drawn from integrated trajectories.

2 Model and conventions

Signature $(-,+,+,+)$, $c = 1$. Interior $f_{\rm in} = 1 - \chi^2 r^2$, exterior $f_{\rm out} = 1 - 2GM/r - \eta^2 r^2$ (here mostly $\eta = 0$), wall $S_{ab} = -\sigma h_{ab}$, $\kappa = 4\pi G\sigma$. With n pointing from interior to exterior, $K_{ab} = e^\mu_a e^\nu_b \nabla_\mu n_\nu$, $[X] = X_+ - X_-$, the Israel conditions $[K_{ab}] - h_{ab}[K] = -8\pi G S_{ab}$ reduce to

$$A_{\rm in} - A_{\rm out} = \kappa R, \qquad A \equiv n^r = \varepsilon\sqrt{f + \dot R^2}, \tag{1}$$

with $K_{\theta\theta} = RA$ and $K^\tau{}_\tau = (\ddot R + \frac{1}{2}f')/A$. Using the identity $A_{\rm in}^2 - A_{\rm out}^2 = f_{\rm in} - f_{\rm out}$,

$$A_{\rm in} = \frac{f_{\rm in} - f_{\rm out} + \kappa^2 R^2}{2\kappa R}, \quad A_{\rm out} = \frac{f_{\rm in} - f_{\rm out} - \kappa^2 R^2}{2\kappa R}, \quad \dot R^2 + V(R) = 0, \quad V = f_{\rm out} - A_{\rm out}^2. \tag{2}$$

Lemma 1. *For $\sigma > 0$, $(R, \varepsilon_{\rm in}, \varepsilon_{\rm out})$ solves the unsquared junction condition iff $\dot R^2 + V = 0$ and $\varepsilon = \mathrm{sgn} A(R)$ on each side. The only spurious solutions of the squared equation are those with $\sigma < 0$.*

With $R = (2GM/L^2)^{1/3} z$, $L^4 = (\chi^2 - \eta^2 + \kappa^2)^2 + 4\kappa^2\eta^2$: $V = 1 - g(z)/E$, $g = z^2 - 2\gamma/z + z^{-4}$, $\gamma = (\chi^2 - \eta^2 - \kappa^2)/L^2$, $E = 4\kappa^2 L^{-8/3}(2GM)^{-2/3}$, and for $\eta = 0$, $\varepsilon_{\rm out} = \mathrm{sgn}(1 - z^3)$, $\varepsilon_{\rm in} = \mathrm{sgn}(1 - \gamma z^3)$.

3 Global structure

For $M < M_{\rm cr} = 4\kappa^3/(G g_{\rm min}^{3/2} L^4)$ there are a bounded branch $(0 \to R_1 \to 0)$ and a bounce branch $(\infty \to R_2 \to \infty)$.

Proposition 2. *(i) $\varepsilon_{\rm out}(R_2) = -1$ always. (ii) $\varepsilon_{\rm out}(R_1) = +1$ iff $E > 2 - 2\gamma$ (regime A). (iii) At $E = 2 - 2\gamma$, $R_1 = 2GM$. (iv) In regime A, the flip radius R_f with $A_{\rm out}(R_f) = 0$ satisfies $R_1 < R_f < R_2$.*

The causal region occupied by the wall is determined by $(\mathrm{sgn} f, \varepsilon, \mathrm{sgn}\dot R)$. We implemented this coordinate-free itinerary and verified it against explicit integration in Kruskal and global de Sitter coordinates: all 15 branches of our atlas agree (Fig. 1). Every inflating branch ends in region III, is invisible from region I, and has a past-singular exterior, in accordance with [2]. The time-symmetric slice at R_2 contains the Einstein–Rosen bridge, a minimal sphere. Hence every classical history through it is singular to the past and to the future, and a transition to it must replace its past.

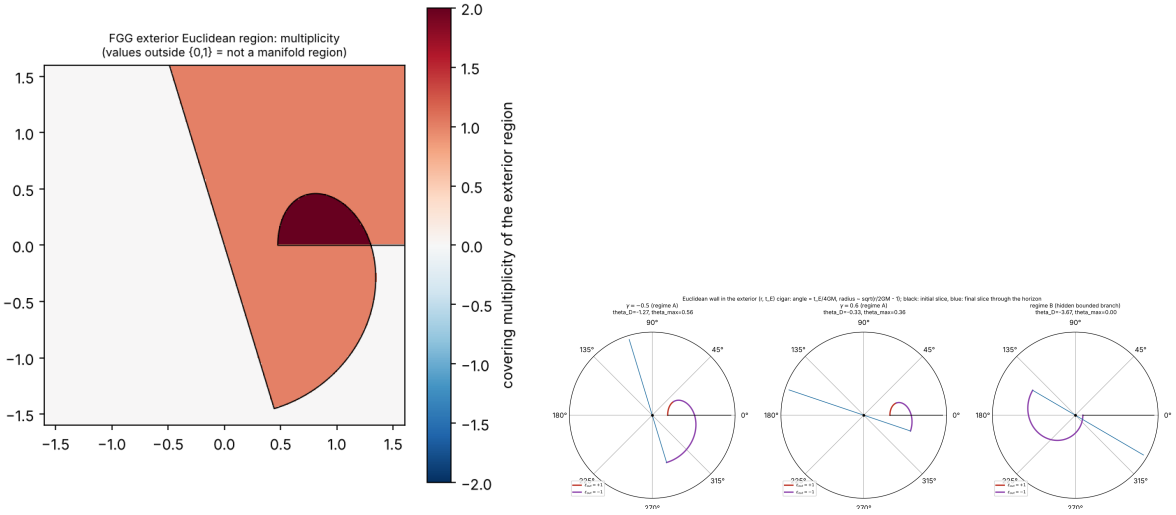


Figure 2: Left: covering multiplicity of the exterior Euclidean region for a regime-A transition. Right: the Euclidean wall in the cigar for three models.

4 The exterior embedding obstruction

Let $\mathcal{C}$ be the (r, t_E) section of Euclidean Schwarzschild (a smooth cigar), with polar coordinates (ρ, θ), $\theta = t_E/4GM$. The Euclidean wall obeys $R'^2 = V$, $f_{\rm out}\, dt_E/d\tau_E = A_{\rm out}(R)$.

Theorem 3 (no manifold instanton)**.** *Suppose that (H1) the half-bounce runs from R_1 to $R_2 > R_1$ with $R' > 0$; (H2) $A_{\rm out}$ has a single simple zero in (R_1, R_2) with $A_{\rm out}(R_1) > 0 > A_{\rm out}(R_2)$; (H3) the initial and final exterior slices are the totally geodesic slices ($\theta = 0$ ray; broken line through the tip at angles $\theta_D, \theta_D + \pi$); (H4) the exterior region is an oriented surface locally isometric to $\mathcal{C}$, with boundary initial slice $\cup$ wall $\cup$ final slice. Then the local isometry is not injective.*

Proof sketch. Orientation places the region outside the $\varepsilon = +1$ arc, inside the $\varepsilon = -1$ arc, counterclockwise of the initial ray, and clockwise of the final ray. If the map were injective, the multiplicity would be 1 on the region, 0 off it, and would jump by one across each boundary piece. Monotonicity of R puts the $\varepsilon = -1$ arc outside the $\varepsilon = +1$ arc. If either arc wraps by $\geq 2\pi$, a radial segment between consecutive windings has multiplicities 0 and 1 at its ends with no crossing between them. Otherwise, at infinity the multiplicity is 1 exactly on the counterclockwise arc $(0, a)$, $a \equiv \theta_D + \pi$, while just outside the $\varepsilon = -1$ arc it is 0. A case analysis in θ_D produces in each case an angle at which these two requirements conflict. The full proof is in the companion file `NO_GO_THEOREMS.md`. □

For the dS|Schwarzschild model, every regime-A transition satisfies (H1)–(H4). Degree-theory multiplicity maps confirm the result: they take values outside $\{0, 1\}$ (Fig. 2), including negative values for small M, where the wall winds dozens of times around the horizon. Regime-B transitions ($\varepsilon_{\rm out} = -1$ throughout) are also non-manifold numerically, with $|\theta_D| > \pi$ over the whole regime. The interior is not obstructed: models with an $\varepsilon_{\rm in}$ flip have embedded interiors, because there the wall approaches the Euclidean horizon.

5 The canonical exponent and its branch

With the asymptotic Killing time as clock and $H = M$, Hamilton's equation $\partial P/\partial M|_R = dt/dR$ gives

$$P = \frac{R}{G}\Big[\operatorname{artanh}\frac{\dot R}{A_{\rm out}} - \operatorname{artanh}\frac{\dot R}{A_{\rm in}}\Big], \qquad 2\sqrt{f_{\rm in}f_{\rm out}}\cosh\frac{GP}{R} = \varepsilon_{\rm in}\varepsilon_{\rm out}(f_{\rm in}+f_{\rm out}-\kappa^2R^2), \quad (3)$$

together with $\partial P/\partial\Lambda_{\rm in}|_{R,M} = -\frac{1}{8\pi G}\frac{4\pi R^3}{3}\frac{dt_{\rm in}}{dR}$ and $\partial P/\partial\sigma|_{R,M} = -4\pi R^2\frac{d\tau}{dR}$. Under the barrier $P = iP_E$, $P_E = \frac{R}{G}(\Theta_{\rm out}-\Theta_{\rm in})$ with $(A,\sqrt{V}) = \sqrt{f}(\cos\Theta,\sin\Theta)$, and $B = 2\int_{R_1}^{R_2} P_E\,dR$.

Proposition 4 (branch selection)**.** *On (R_1,R_2) we have $f \geq V > 0$ on both sides, so P is analytic there. Its branch points are the turning points and the horizons $f = 0$; the flip radius is a regular point. The WKB phase is the analytic continuation along the barrier, $\Theta = \operatorname{atan2}(\sqrt{V},A) \in [0,\pi]$. The principal branch differs by a horizon monodromy ($P \to P \pm i\pi R/G$, $t \to t \pm i\beta/2$), which is the Euclidean half-turn slicing jump of FGG. The principal branch gives $B < 0$ in 41 of 45 test models.*

6 Closed form at $M = 0$

Theorem 5. *Let $M = 0$, $R_0 = 2\kappa/L^2$, $p = -A_{\rm out}(R_0)$, $q = A_{\rm in}(R_0)$ ($p+q = \kappa R_0 > 0$). Then*

$$S_E = -\frac{\pi R_0^2}{2G}\Big[\frac{1}{1+p}+\frac{1}{1+q}\Big], \qquad B = \frac{\pi R_0^2}{2G}\frac{p+q+2p_+q_+}{(1+|p|)(1+|q|)} = S_E + \frac{1}{4G}\Big(\sum A_{\max} - \sum A_{\min}\Big), \quad (4)$$

the sums running over the extremal spheres of the instanton's time-symmetric slice. If $q < 0$, then $B = B_{\rm CdL}$, the decay exponent of the false vacuum.

The proof uses $J(s) = \int_0^1 x\operatorname{atan2}(\sqrt{1-x^2},sx)dx = \pi/4(1+s)$ ($s \geq 0$), $\pi/2 - \pi/4(1-s)$ ($s < 0$), together with the cap volumes. Gravitational $O(4)$ bounces of a quartic scalar converge to $B_{\rm CdL}$ as $O(w)$ in the wall width (0.18% at $w = 0.005$).

7 First law and Smarr identity

Theorem 6. *Let $\tilde B = B - \mathcal{A}/4G$, where $\mathcal{A}$ collects the endpoint areas ($\pm 4\pi R_i^2$ where the wall is an extremal sphere of its slice, plus $4\pi/\chi^2$ if the final slice contains the de Sitter equator). Then, within a regime,*

$$d\tilde B = \frac{\mathrm{Vol}_E^{\rm in}}{8\pi G}d\Lambda_{\rm in} + A_E^W\,d\sigma - \Delta t_E^W\,dM, \qquad B = -\frac{\Lambda_{\rm in}}{8\pi G}\mathrm{Vol}_E^{\rm in} - \frac{\sigma}{2}A_E^W - \frac{M}{2}\Delta t_E^W + \frac{\mathcal{A}}{4G}, \quad (5)$$

where $\mathrm{Vol}_E^{\rm in}$ is the Euclidean interior 4-volume of the full bounce, A_E^W the wall's Euclidean area, and Δt_E^W the exterior Euclidean Killing time along the wall.

Proof. The first law is the Euclidean continuation of the three Hamilton–Jacobi relations, with endpoint contributions $2P_E(R_i)dR_i = d(\mathcal{A}/4G)$. The Smarr identity follows from Euler's relation for $\tilde B$, which is homogeneous of degree 2 under $(\Lambda,\sigma,M) \to (\lambda^{-2}\Lambda,\lambda^{-1}\sigma,\lambda M)$. □

The first two terms are the on-shell action of the interior region and the wall. The exterior pseudomanifold contributes only $-\frac{M}{2}\Delta t_E^W$, which tends to zero as $M \to 0$ even though the Euclidean winding diverges. Integrating the first law in M from the $M = 0$ instanton gives $B(M) = S_E^{(0)} + \mathcal{A}(M)/4G - \int_0^M \Delta t_E^W\,dM'$. Numerically the identity holds to 2.5×10^{-8} over 45 models.

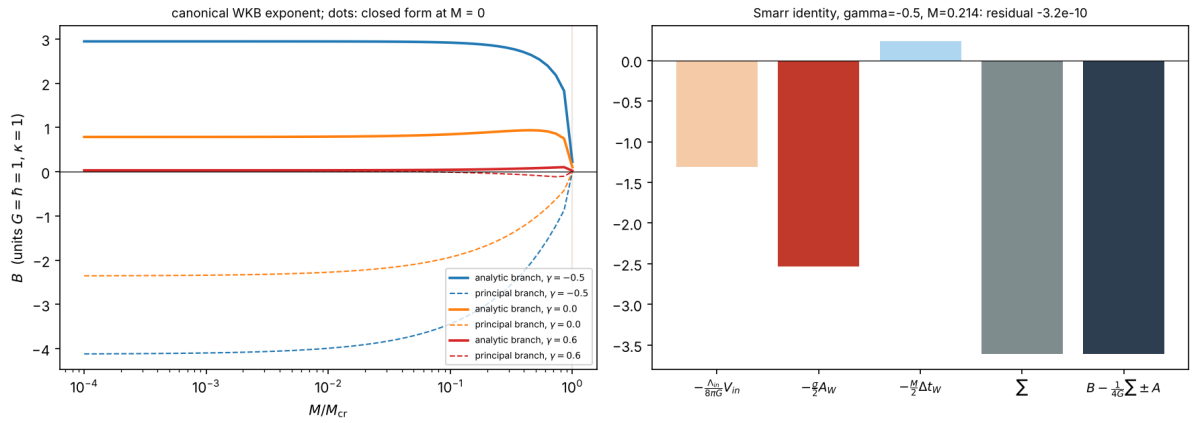


Figure 3: Left: $B(M)$ for the analytic (solid) and principal (dashed) branches; dots: Theorem 5. Right: the terms of the Smarr identity.

8 Numerical results

Over regime A (laboratory data) the ratio $B/B(M \to 0)$ is bounded below by 0.56 for all $\gamma \in [-0.95, 0.95]$, and by 1 for $\gamma \gtrsim 0.2$, where $B(M \to 0) = B_{\mathrm{CdL}}$ (Fig. 3). For GUT-like parameters, $\sigma^{1/3} = \rho^{1/4} = 10^{16}\,\mathrm{GeV}$ ($M_{\mathrm{cr}} \approx 5.6\,\mathrm{g}$), $B = 133$ only for picogram masses, while $B \approx 7 \times 10^4$ at $10^{-3} M_{\mathrm{cr}}$ and 4×10^6 at $0.5 M_{\mathrm{cr}}$.

9 Discussion and limitations

The three obstructions to universe creation in an asymptotically flat parent, classical (Farhi–Guth), initial-data (bag-of-gold minimal sphere) and semiclassical (Theorem 3), share one origin: the creation of a closed marginally trapped surface in the exterior. Within the reduced canonical theory, the exponent is nevertheless unambiguous and covariantly expressible. Open problems: the exact quantum amplitude (an exact solution is possible at $M = 0$, where the constraint becomes an ODE in the momentum representation); whether Lorentzian path integrals select the same contour; prefactors; thick-wall versions of Theorems 3 and 6; non-spherical perturbations. Our novelty claims are provisional pending comparison with the full texts of [3, 4, 7, 8].

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